

WSL Phase 8 — rev-B Controller Board — Pre-Power Test Checklist

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Pre-Power Bench Test Checklist (shorts · continuity · first power)

Field	Value
Board ID / serial	_____
Date	_____
Tester	_____
DMM model	_____
Bench supply	_____ V set, _____ mA current limit

Derived from the live board netlist (`ws1-phase8b.routed-manual.kicad_pcb`). Every probe point below is a real net/pad on this board. Do the phases **in order**. Do **not** apply power until Phase 0–2 pass.

How to use the meter

Short checks (Phase 1): continuity/beep + low- Ω range. “NOT shorted” = no steady beep, reads high. Caps charge through the meter, so rail-to-rail readings *climb* as C11/decoupling fill — that’s normal; a real short sits steady near 0 Ω .

Continuity (Phase 2): continuity/beep. “Beep” = $\sim 0\ \Omega$ = good joint / same net. Probe the **pin metal** (top contact or back-side tail), not the plastic.

Diode mode where noted (D17).

Reference grounds: logic measurements → **TP2 (GND)**. Field-domain measurements → **TP5 (FIELD_GND)**. **Never bridge a meter lead across TP2 and TP5** — that defeats the U37 isolation barrier.

Test-point locations (silk labels TP1–TP16)

TP	Net	TP	Net
TP1	VCC_5V (logic +5 V, post-D17)	TP9	WDOG_TIMING_NODE
TP2	GND (logic ground)	TP10	NE555_TRIG
TP3	VCC_3V3 (Pico regulator)	TP11	NE555_OUT
TP4	FIELD_WET_V (isolated +5 V)	TP12	WDOG_OK_PULLDOWN
TP5	FIELD_GND (isolated ground)	TP13	ARM_PERMIT
TP6	I2C_SDA	TP14	RP2040_OK
TP7	I2C_SCL	TP15	SAFE_STOP_RETURN
TP8	WDOG_KICK	TP16	RELAY_ENABLE_RAIL

STOP CONDITIONS (any phase): rail-to-ground short · ANY FIELD_GND ↔ GND continuity · current pegging the limit at power-up · any part heating. Power off and find the cause before continuing.

PHASE 0 — Visual inspection (no meter, both sides, magnification)

Bridges / defects - ☐ No solder balls, whiskers, or flux bridges anywhere — **top side** - ☐ No solder balls, whiskers, or flux bridges anywhere — **bottom side** - ☐ **A1 Pico** castellations — no bridges between adjacent castellations - ☐ **3× MCP23017** SOIC leads (U1/U2/U3) — no lead-to-lead bridges - ☐ **32-opto bank** (PC817 U4–U35) — no pin bridges - ☐ **Relay pins** K1–K6 — no bridges - ☐ **U37 (TMA-0505S)** 4 pins — no bridges

Cold / insufficient joints (dull, balled, not wicked through) - ☐ All connector joints inspected — full, shiny fillets - ☐ **J13 reseated pins** specifically — confirmed full fillets both sides

Polarized / orientation — HAND-SOLDERED (where errors happen) - ☐ **A1 Pico** — pin-1 / USB edge matches silk - ☐ **U37 (TMA-0505S)** — square pad = pin 1 = +Vin (verified pinout); body flush

Polarized / orientation — JLC-PLACED (glance for tombstone/rotation/wrong part) - ☐ **D17 (SS14)** reverse-protection Schottky — cathode band toward the rail (silk) - ☐ **D15, D16 (1N4148)** watchdog diodes — cathode bands match silk - ☐ **D1, D3, D5, D7, D9, D11, D13 (1N4148)** relay flyback diodes — cathode bands match silk - ☐ **C11 electrolytic** ("100 16V") — minus stripe matches silk polarity - ☐ **K1–K6** relays all seated flat; **K7 (M1) is EMPTY** (DNP — correct) - ☐ **D2/D4/D6/D8/D10/D12/D14 (MOV)** footprints are **EMPTY** (DNP — correct)

Mechanical - ☐ Every connector body flush and square — no lifted/tilted headers

PHASE 1 — Unpowered short check · SUPPLY DISCONNECTED

Meter on continuity/low- Ω . These are the checks that prevent a power-on disaster.

- ☐ **TP1 (VCC_5V) ↔ TP2 (GND) — NOT shorted** (high Ω). Steady $\sim 0\ \Omega \rightarrow$ **STOP** (5 V ↔ gnd bridge)
 - ☐ **TP3 (VCC_3V3) ↔ TP2 (GND) — NOT shorted**. Steady $\sim 0\ \Omega \rightarrow$ **STOP**
 - ☐ **TP1 (VCC_5V) ↔ TP3 (VCC_3V3) — NOT shorted** (3V3 is regulated from 5 V, not tied)
 - ☐ **J2 pin 1 (VCC_5V_RAW) ↔ TP2 (GND) — NOT shorted** (input side, before D17)
 - ☐ **TP4 (FIELD_WET_V) ↔ TP5 (FIELD_GND) — NOT shorted** (U37 isolated output)
 - ☐ **TP5 (FIELD_GND) ↔ TP2 (GND) — OPEN / no continuity ← isolation barrier**. ANY beep $\rightarrow$ **STOP**
 - ☐ **TP4 (FIELD_WET_V) ↔ TP2 (GND) — OPEN** (field/logic isolation). ANY beep $\rightarrow$ **STOP**
 - ☐ **TP4 (FIELD_WET_V) ↔ TP1 (VCC_5V) — OPEN** (field/logic isolation). ANY beep $\rightarrow$ **STOP**
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PHASE 2 — Continuity / joint integrity · still unpowered

2A — Logic ground common (probe TP2 GND ↔ each; all should beep)

- ☐ TP2 ↔ **J2 pin 2 (GND)**
- ☐ TP2 ↔ **J2 pin 3 (GND)**
- ☐ TP2 ↔ **J1 pin 2 (GND)**
- ☐ TP2 ↔ **J1 pin 12 (GND)**
- ☐ TP2 ↔ **J13 pin 2 (GND)**
- ☐ TP2 ↔ **C11 minus (–) terminal**

2B — Logic +5 V (probe TP1 VCC_5V ↔ each; all should beep)

- ☐ TP1 ↔ **J1 pin 1 (VCC_5V)**
- ☐ TP1 ↔ **J13 pin 1 (VCC_5V)**
- ☐ TP1 ↔ **J14 pin 1 (VCC_5V)**

2C — Input rail + reverse-protection diode D17

- ☐ **J2 pin 1 (VCC_5V_RAW) ↔ D17 anode pad** — beep (same raw-input net)
- ☐ **D17 in diode mode:** J2 pin 1 (anode) $\rightarrow$ TP1 (cathode) reads **$\sim 0.3\ \text{V}$ forward, OPEN reversed** — confirms D17 present and **correctly oriented**
- ☐ Note: J2 pin 1 $\leftrightarrow$ TP1 is **not** a plain beep — it passes through D17 (expected)

2D — Logic 3.3 V and named control signals (probe TP ↔ J1 pin; should beep)

- ☐ **TP3** (VCC_3V3) ↔ **J1 pin 11**
- ☐ **TP6** (I2C_SDA) ↔ **J1 pin 3**
- ☐ **TP7** (I2C_SCL) ↔ **J1 pin 4**
- ☐ **TP8** (WDOG_KICK) ↔ **J1 pin 7**
- ☐ **TP13** (ARM_PERMIT) ↔ **J1 pin 8**
- ☐ **TP14** (RP2040_OK) ↔ **J1 pin 13**
- ☐ **J1 pin 5** (PI_UART_TX) — joint solid (no dedicated TP; ring to Pico TX pad or verify visually + wiggle)
- ☐ **J1 pin 6** (PI_UART_RX) — joint solid (ring to Pico RX pad or verify visually + wiggle)
- ☐ **J1 pin 9** (MCP_INT_A) — joint solid (ring to MCP INTA pad or verify visually + wiggle)
- ☐ **J1 pin 10** (MCP_INT_B) — joint solid (ring to MCP INTB pad or verify visually + wiggle)

2E — Field ground common (probe **TP5** FIELD_GND ↔ each; FIELD domain — all beep)

- ☐ TP5 ↔ **J3 pin 9** (FIELD_GND)
- ☐ TP5 ↔ **J3 pin 10** (FIELD_GND)
- ☐ TP5 ↔ **J4 pin 14** (FIELD_GND)
- ☐ TP5 ↔ **J5 pin 12** (FIELD_GND)
- ☐ Confirm all four also read **OPEN to TP2** (already covered in Phase 1 — re-verify any that didn't)

2F — J13 lamp-return pins → their 330 Ω resistors (the row of “331” 0805s by J13)

- ☐ **J13 pin 3** (LED_L_FIRST_RETURN) ↔ **R90** — beep one pad / ~330 Ω other pad
- ☐ **J13 pin 4** (LED_L_SECOND_RETURN) ↔ **R93** — beep / ~330 Ω
- ☐ **J13 pin 5** (LED_L_STRIKE_RETURN) ↔ **R96** — beep / ~330 Ω
- ☐ **J13 pin 6** (LED_L_FOUL_RETURN) ↔ **R99** — beep / ~330 Ω
- ☐ **Wiggle test** on the two reseated J13 pins while probing — reading must stay solid

2G — J14 safety loop

- ☐ **J14 pin 1** (VCC_5V) ↔ **TP1** — beep
- ☐ **J14 pin 2** ↔ **J14 pin 3** — **beep** (bridged on-board, SAFE_TBSC_RETURN)
- ☐ **J14 pin 4** (SAFE_STOP_RETURN) ↔ **TP15** — beep
- ☐ **J14 pin 1** ↔ **pin 2** — **OPEN** (loop open until an external NC contact closes it)
- ☐ **J14 pin 3** ↔ **pin 4** — **OPEN** (second external loop)

2H — Motion-output terminals (relay dry contacts; J12/M1 is DNP — skip)

- ☐ **J6** (K1, S): both terminals solid joint to relay K1; no bridge to J7
- ☐ **J7** (K2, T): both terminals solid joint to K2; no bridge to neighbors
- ☐ **J8** (K3, SP): both terminals solid joint to K3; no bridge to neighbors
- ☐ **J9** (K4, BE): both terminals solid joint to K4; no bridge to neighbors
- ☐ **J10** (K5, M): both terminals solid joint to K5; no bridge to neighbors
- ☐ **J11** (K6, M2): both terminals solid joint to K6; no bridge to J10
- ☐ Note: pin 1↔pin 2 of each = relay contact (open or closed depending on G5LE NO/NC at rest) — **not** a fault by itself

2I — Adjacent-pin short scan (drag probes across neighbors; OPEN except noted ties)

- ☐ **J1** (20-pin) — all adjacent pairs OPEN (no two used pins share a net)
- ☐ **J2** — pin 2↔pin 3 **beep** (both GND, OK); pin 1↔pin 2 OPEN
- ☐ **J3** (10-pin) — all adjacent OPEN **except** pin 9↔pin 10 beep (both FIELD_GND, OK)
- ☐ **J4** (14-pin) — all adjacent pairs OPEN
- ☐ **J5** (12-pin) — all adjacent pairs OPEN
- ☐ **J13** (6-pin) — all adjacent pairs OPEN
- ☐ **J14** (4-pin) — pin 2↔pin 3 beep (OK); all other adjacent pairs OPEN

PHASE 3 — First power-up (current-limited)

- ☐ Bench supply set: **5.0 V** (or ~5.3 V for full rail margin), **current limit ~200 mA**
 - ☐ Connect supply to **J2** — **pin 1 = +5 V, pins 2/3 = GND**. Double-check polarity before switching on
 - ☐ Switch on. **Steady current < ~100 mA** (Pico + 3× MCP + NE555 + pull-ups). Pegging the limit → **STOP**. → Record actual: _____ mA
 - ☐ **TP1 (VCC_5V) ≈ 4.6 V** (= 5 V – SS14 drop; expected). → Record: _____ V
 - ☐ **TP3 (VCC_3V3) ≈ 3.3 V** (confirms Pico is alive and regulating). → Record: _____ V
 - ☐ **TP4 → TP5 (FIELD_WET_V) ≈ +5 V isolated** (measure to TP5, not TP2). → Record: _____ V
 - ☐ **TP5 ↔ TP2 still OPEN under power** (isolation holds live)
 - ☐ **No part hot** — finger-touch A1, U37, U1/U2/U3 (MCP), the NE555, the FETs, K1–K6. Anything more than mildly warm → **STOP, power off**
 - ☐ **TP16 (RELAY_ENABLE_RAIL)** — record state (likely held LOW until firmware drives the watchdog kick + ARM): _____ V
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RESULT

- ☐ **PASS** — all phases clean; board cleared for functional bring-up (I²C enumerate MCPs @ 0x20/0x21/0x22, relay-click, watchdog rail-drop)
- ☐ **HOLD** — issue(s) found (note below)

Notes / anomalies:

Tester signature	_____
Date / time	_____